Table of Contents

1. **Selected Interactive System:** 2
   - **1.1 System Description:** 2
2. **Usability Test Planning:** 2
   - **2.1 Usability Goals and Objectives:** 2
   - **2.2 Target User Characteristics:** 3
   - **2.3 Tasks and Scenarios:** 3
   - **2.4 Test Environment and Tools:** 4
   - **2.5 Evaluation Metrics and Success Criteria:** 4
3. **Usability Evaluation Method:** 4
4. **Usability Evaluation Results:** 5
5. **Interface Redesign and Improvements:** 10
   - **5.1 Overview of Redesign Approach:** 10
   - **5.2 Redesigned Components Analysis:** 11
   - **5.3 Usability Principles Applied:** 19
6. **Discussion:** 20
   - **6.1 Comparison of Original and Redesigned Interfaces:** 20
   - **6.2 Summary:** 21

1. Selected Interactive System:

System Name: "RABTA" (Pakistan Railways E-Ticketing System).
URL: www.pakrailways.gov.pk

1.1 System Description:

The selected interactive system for this study is **RABTA**, the official web-based e-ticketing and information portal of Pakistan Railways. The system allows users to search train schedules, create and manage accounts, book railway tickets, and handle various travel related activities online. Since RABTA is a national public sector service, it is designed for a wide and diverse audience, including users of different age groups, educational levels, and technical skills.

RABTA includes several important interactive elements, such as account registration forms, login screens, train search filters, booking tables, and verification processes. Because it serves as a critical public utility and involves multiple user-facing workflows, RABTA represents a relevant and practical case for conducting a usability evaluation.

2. Usability Test Planning:

This section outlines the structured approach used to evaluate the usability of the RABTA system. The planning phase ensures that the evaluation is focused, systematic, and aligned with usability engineering principles.

2.1 Usability Goals and Objectives:

The main objective of this evaluation is to examine the learnability, efficiency, accessibility, and error prevention features of the RABTA system, with a particular focus on new users, elderly individuals, and users with limited technical skills**.**

The key usability goals of the evaluation are outlined below:

Goal 1: (Accessibility)

This goal aims to determine whether the interface is visually clear and easy to navigate for elderly users. Special attention is given to factors such as text readability, colour contrast, spacing, and overall visual hierarchy.

Goal 2: (Error Prevention)

This goal focuses on evaluating how effectively the system helps users avoid mistakes and how well it supports error recovery during important tasks, such as account registration and verification.

Goal 3: (Clarity and Understandability)

This goal assesses whether the system's navigation labels, terminology, and system messages are easy to understand for average users, especially those with limited technical knowledge.

2.2 Target User Characteristics:

The target users of the RABTA system include the general population of Pakistan. For the purpose of this usability evaluation, the following user profile has been considered.

i. **Demographics:** Pakistani citizens ranging from 18 years of age to 65 and above.
ii. **Technical Proficiency:** Users with novice to intermediate technical skills, including individuals who have limited experience with online forms and digital payment systems.
iii. **Devices:** Users primarily access the system through desktop computers and mobile web browsers.
iv. **Language Proficiency:** Many users have Urdu or a regional language as their primary language and may face difficulty understanding English-based interfaces, especially when technical or institutional terms are used.

Due to the diversity of this user group, usability plays a crucial role in ensuring that users can complete tasks successfully and without confusion.

2.3 Tasks and Scenarios:

The evaluation focused on three core tasks that reflect the main functionalities of the RABTA system. These tasks were chosen to represent both basic entry-level interactions and goal-oriented user activities.

i. **Account Creation:**

   Scenario: A user registers a new account by providing a CNIC number and mobile phone number, followed by completing the email verification process.

ii. **System Login:**

   Scenario: The user logs in to an existing account in order to access account-related features.

iii. **Search and Booking Exploration:**

   Scenario: The user searches for a train traveling from Rawalpindi to Quetta on a selected date and checks the availability of seats.

Together, these tasks represent a complete user journey, starting from initial onboarding and ending with successful goal completion.

2.4 Test Environment and Tools:

i. **Environment:**

The usability evaluation was carried out in a standard desktop environment using Windows 10/11 with the Google Chrome web browser. This setup closely represents a typical home-user configuration and reflects how most users are likely to access the system.

ii. **Evaluation Tool:**

The evaluation was primarily conducted using the **Heuristic Evaluation** method, based on expert inspection of the interface. Screen capture tools were used to record interface elements and document any usability issues observed while performing the selected tasks.

2.5 Evaluation Metrics and Success Criteria:

i. **Task Success Rate:**

These measures whether users are able to complete each task successfully without facing system-blocking errors.

ii. **Error Frequency:**

This examines how often users misunderstand labels, enter incorrect information, or experience errors caused by unclear or poor interface design.

iii. **Subjective Satisfaction:**

This considers users' perceived ease of use, clarity of the interface, trustworthiness of the system, and overall readability and understandability.

3. Usability Evaluation Method:

Method Selected: "Heuristic Evaluation".

Heuristic Evaluation was chosen as the usability evaluation method because it is effective in identifying interface-level design issues using well-established usability principles. The RABTA system was evaluated using **Nielsen's 10 Usability Heuristics**, which allowed for a systematic analysis of problems related to navigation, consistency, error handling, clarity, and accessibility.

This method is particularly suitable for evaluating the RABTA system for several reasons. First, it does not require a large number of participants, making it practical to conduct. Second, it is highly effective for early-stage and diagnostic evaluations, where the goal is to identify

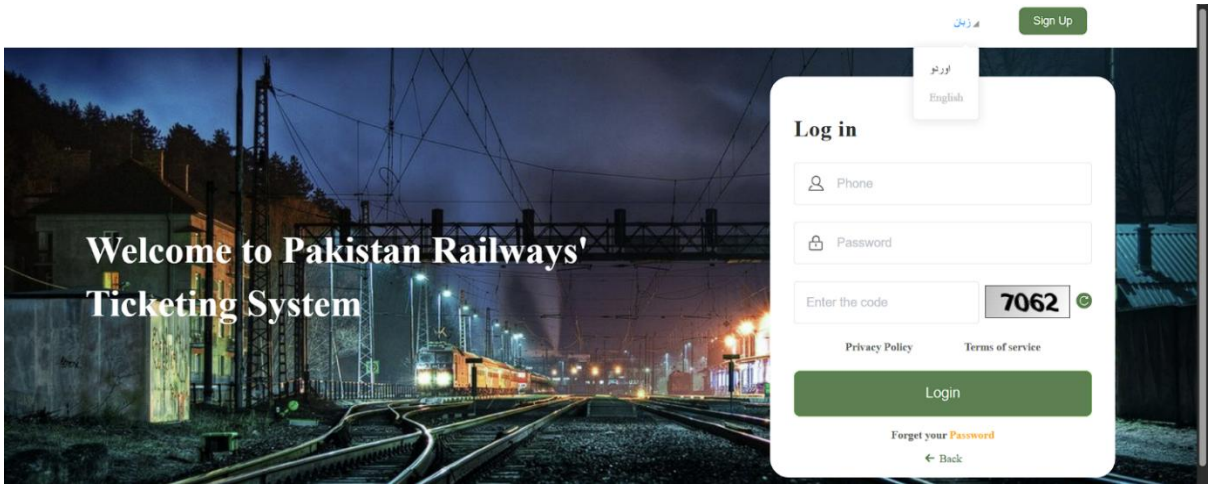
usability problems rather than measure performance quantitatively. Finally, it helps uncover issues that can be directly addressed through interface improvements and redesign.

Overall, this approach aligns well with the inspection-based nature of the usability issues identified in the system.

4. Usability Evaluation Results:

This section presents the usability issues identified during the heuristic evaluation. Each issue is mapped to the relevant usability heuristic and includes an explanation of its impact on user experience.

Issue 1: (Poor Visibility and Contrast on Log in page)



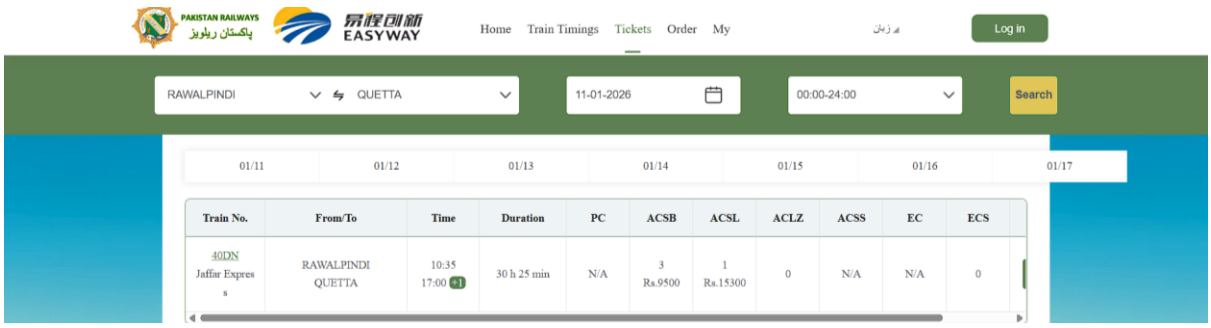
Problem:
The login input fields use a white background that blends with the surrounding page, making them difficult to visually distinguish. In addition, the header and footer do not have enough visual separation from the main content.

Heuristics Violated:

- Visibility of System Status.
- Aesthetic and Minimalist Design.

Impact:
This design issue makes it harder for elderly users and users with visual impairments to easily identify interactive fields. As a result, accessibility is reduced and task completion time is increased.

Issue 2: (Use of Internal Railway Jargon on Search Results Page)



Problem:

The search results table displays abbreviations such as ACSB, ACSL, ACLZ, and PC without providing any explanation or clarification.

Heuristic Violated:

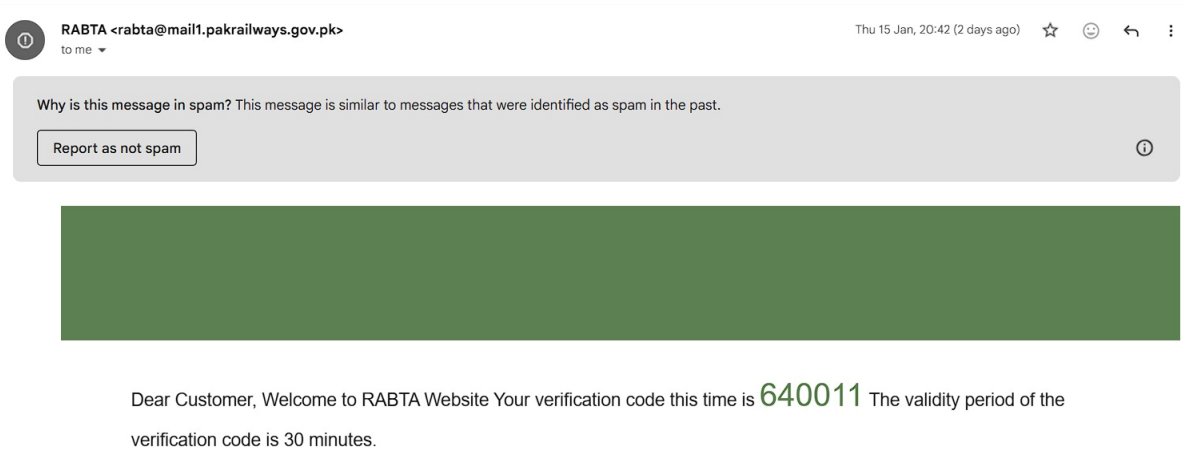
- Match Between System and the Real World.

Impact:

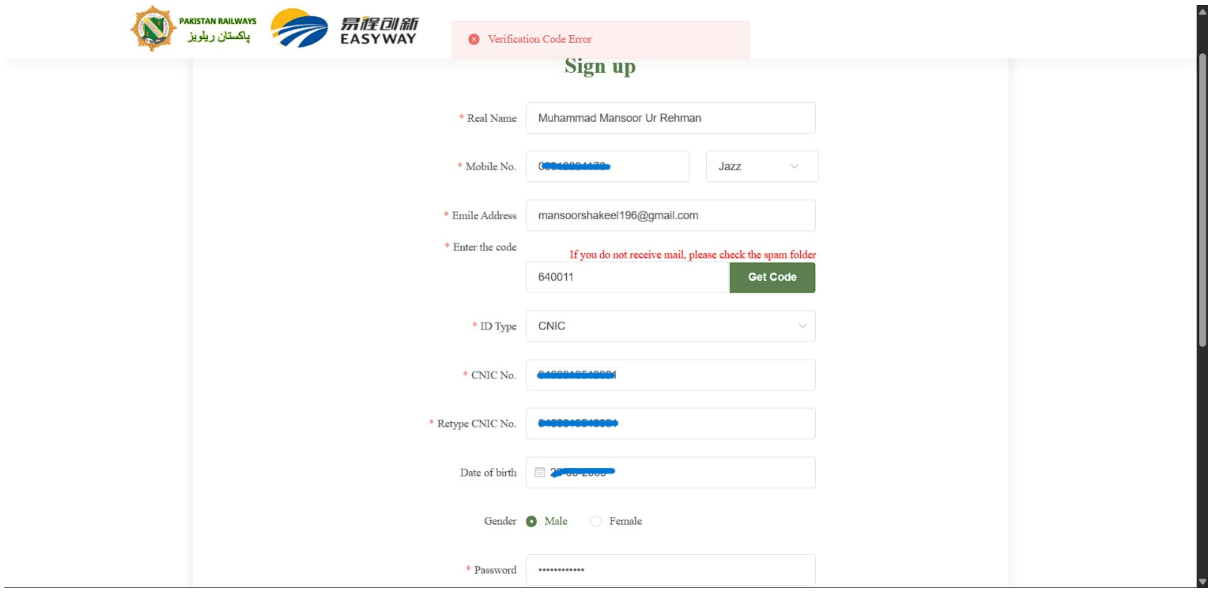
By using internal railway terminology instead of user-friendly language, the system creates confusion for users. This can lead to misunderstandings and increases the risk of incorrect booking decisions.

Issue 3: (Critical Functionality Failure During Signup and Verification)

Email:



Sign-up Page:



Problem:
The email verification process is inconsistent. Users often receive an "Invalid Code" error even after entering the correct verification code, and in some cases, verification prompts appear before the account creation process is fully completed.

Heuristics Violated:

- Error Prevention.
- Help Users Recognize, Diagnose, and Recover from Errors.

Impact:
This issue prevents new users from successfully completing the registration process, effectively blocking access to the system and causing frustration for first-time users.

Issue 4: (Ambiguous Navigation Labels on Homepage)

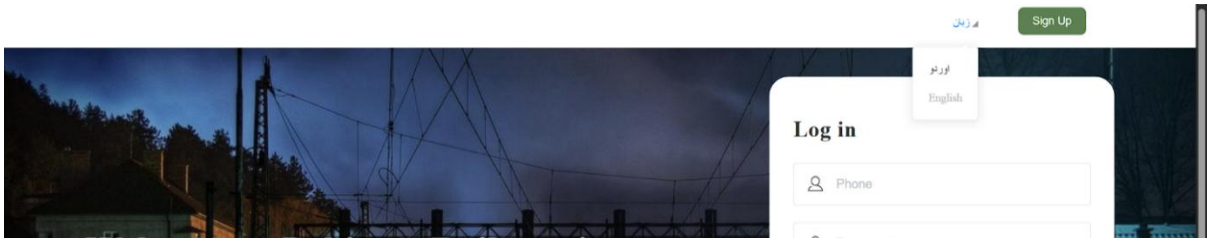
Problem:
Navigation items such as "My" and "Order" are not descriptive and fail to clearly indicate their purpose or destination.

Heuristic Violated:

- Recognition Rather Than Recall.

Impact:
Users have to guess the meaning of these labels, which increases cognitive load and reduces navigation efficiency across the system.

Issue 5: (Limited Cultural and Linguistic Localization)



Problem:
The interface supports only English and Urdu, leaving out major regional languages such as Punjabi, Pashto, and Sindhi.

Heuristic Violated:

- Match Between System and the Real World.

Impact:
A large portion of users may feel excluded or find the system difficult to use, reducing accessibility and overall user satisfaction.

Issue 6: (Hidden Primary Action Booking Button)

Train No.	From/To	Time	Duration	PC	ACSB	ACSL	ACLZ	ACSS	EC	ECS
101UP Subak Raftar	GUJRANWALA RAWALPINDI	08:00 11:55	3 h 55 min	0	N/A	N/A	3 Rs.1950	38 Rs.1850	N/A	164 Rs.1000
7UP Tezgam	GUJRANWALA RAWALPINDI	15:30 20:00	4 h 30 min	N/A	0	3 Rs.3100	13 Rs.2250	N/A	45 Rs.1200	55 Rs.1150
103UP Subak Kharram	GUJRANWALA RAWALPINDI	17:35 21:40	4 h 5 min	1 Rs.2250	N/A	N/A	1 Rs.1950	11 Rs.1850	N/A	298 Rs.1000
1UP Khyber Mail	GUJRANWALA RAWALPINDI	22:15 02:15 +1	4 h	N/A	0	0	1 Rs.2000	N/A	29 Rs.850	49 Rs.800

From/To	Time	Duration	PC	ACSB	ACSL	ACLZ	ACSS	EC	ECS	Remark
GUJRANWALA RAWALPINDI	08:00 11:55	3 h 55 min	0	N/A	N/A	3 Rs.1950	38 Rs.1850	N/A	164 Rs.1000	Booking
GUJRANWALA RAWALPINDI	15:30 20:00	4 h 30 min	N/A	0	3 Rs.3100	13 Rs.2250	N/A	45 Rs.1200	55 Rs.1150	Booking
GUJRANWALA RAWALPINDI	17:35 21:40	4 h 5 min	1 Rs.2250	N/A	N/A	1 Rs.1950	11 Rs.1850	N/A	298 Rs.1000	Booking
GUJRANWALA RAWALPINDI	22:15 02:15 +1	4 h	N/A	0	0	1 Rs.2000	N/A	29 Rs.850	49 Rs.800	Booking

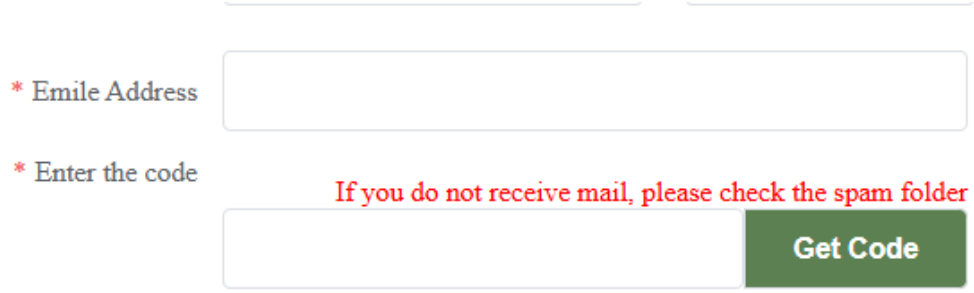
Problem:
On the train search results page, the main "Booking" button is positioned off-screen, forcing users to scroll horizontally to access it.

Heuristic Violated:

- Flexibility and Efficiency of Use.

Impact:
Users may assume that booking is unavailable, which can lead to task failure or abandonment of the process.

Issue 7: (Spelling Errors in Critical Forms)



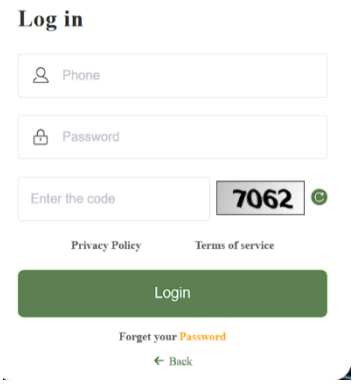
Problem:
The signup form contains a spelling mistake, with "Email Address" incorrectly written as "Emile Address."

Heuristic Violated:

- Aesthetic and Minimalist Design.

Impact:
Spelling errors like this can reduce user trust in the system, particularly during sensitive tasks that involve personal or payment information.

Issue 8: (Misplaced Legal Information)



Problem:
Links to the Terms of Service and Privacy Policy are placed in non-standard locations, making them hard to find.

Heuristic Violated:

- Consistency and Standards.

Impact:
Users may have difficulty locating important legal information, which can reduce transparency and diminish trust in the system.

5. Interface Redesign and Improvements:

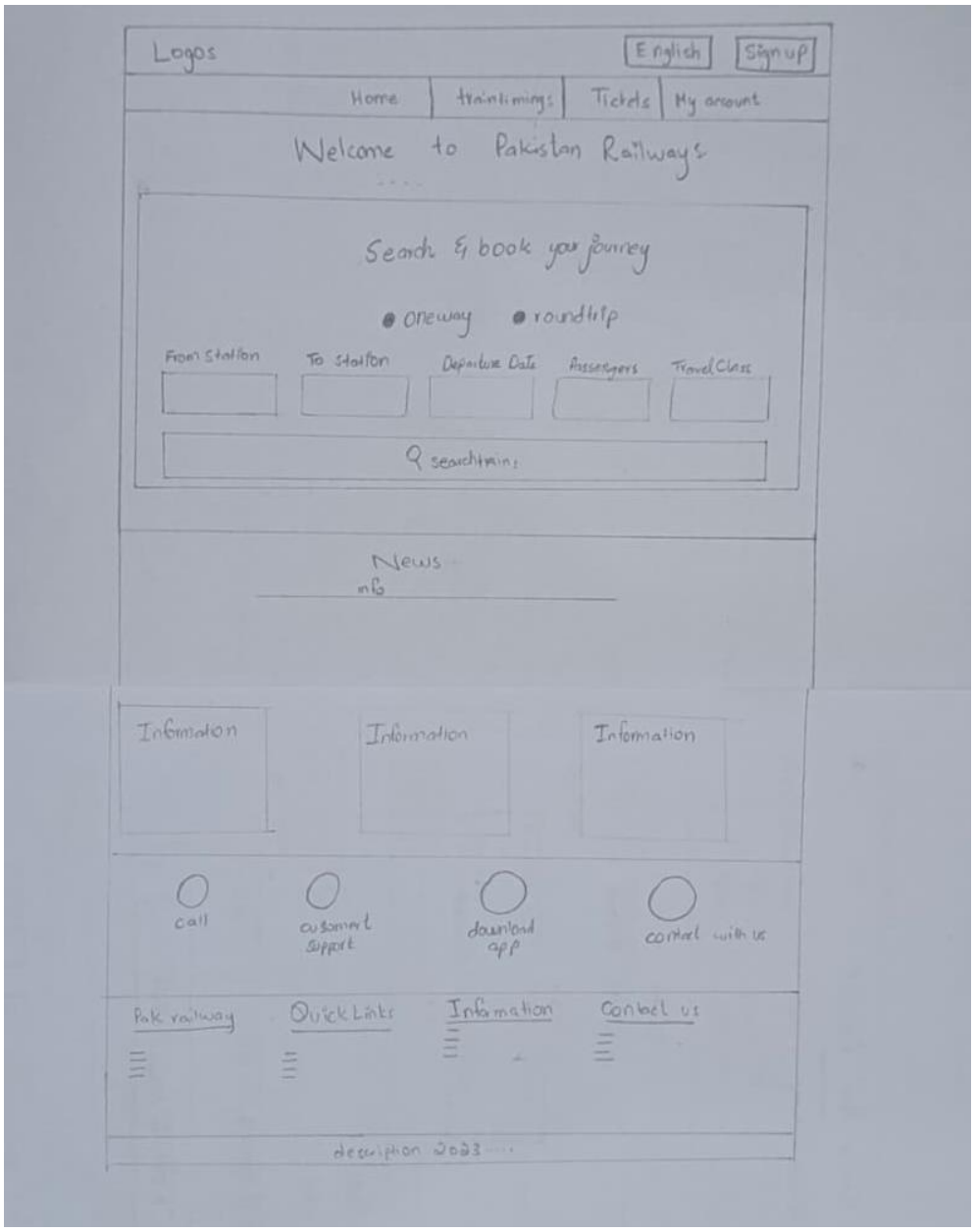
5.1 Overview of Redesign Approach:

The redesigned interfaces address the usability issues identified in the heuristic evaluation through a user-centered design approach. The improvements focus on accessibility, clarity, consistency, error prevention, and cultural localization. Each redesigned page maintains a consistent visual identity with the Pakistan Railways brand while implementing modern UI/UX principles.

5.2 Redesigned Components Analysis:

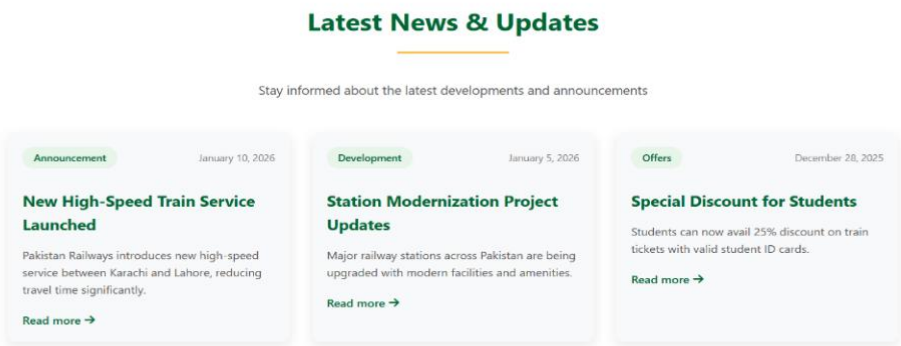
i. Home Page:

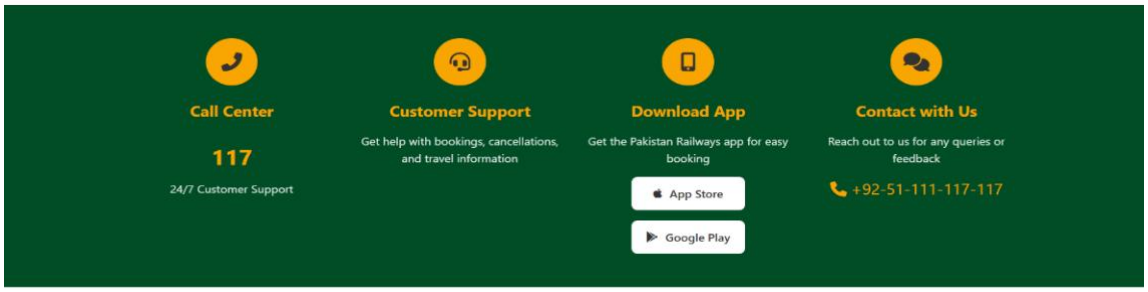
Lo-Fidelity:

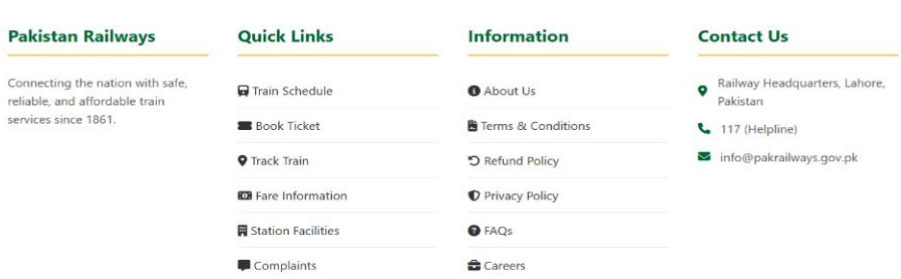


Hi-Fidelity:

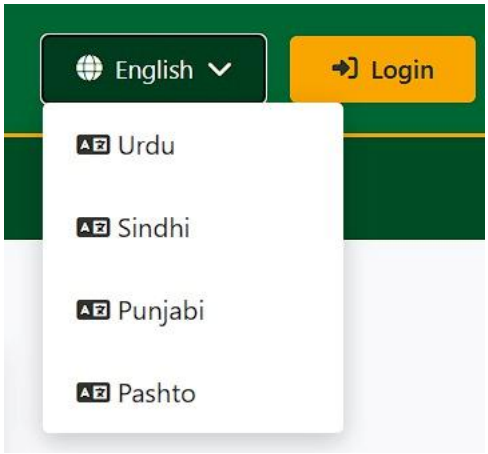
EASYWAY
English
Login
Home
Train Timings
Tickets
My Account
Welcome to Pakistan Railways
Book your train tickets online with ease and convenience
Search & Book Your Journey
One Way
Round Trip
From Station
To Station
Departure Date
Passengers
Travel Class
Select departure static
Select arrival station
1/16/2026
1
Select class
Search Trains


Latest News & Updates
Stay informed about the latest developments and announcements
Announcement
January 10, 2026
New High-Speed Train Service Launched
Pakistan Railways introduces new high-speed service between Karachi and Lahore, reducing travel time significantly.
Read more →
Development
January 5, 2026
Station Modernization Project Updates
Major railway stations across Pakistan are being upgraded with modern facilities and amenities.
Read more →
Offers
December 28, 2025
Special Discount for Students
Students can now avail 25% discount on train tickets with valid student ID cards.
Read more →


Call Center
117
24/7 Customer Support
Customer Support
Get help with bookings, cancellations, and travel information
Download App
Get the Pakistan Railways app for easy booking
App Store
Google Play
Contact with Us
Reach out to us for any queries or feedback
+92-51-111-117-117


Pakistan Railways
Connecting the nation with safe, reliable, and affordable train services since 1861.
Quick Links
Train Schedule
Book Ticket
Track Train
Fare Information
Station Facilities
Complaints
Information
About Us
Terms & Conditions
Refund Policy
Privacy Policy
FAQs
Careers
Contact Us
Railway Headquarters, Lahore, Pakistan
117 (Helpline)
info@pakrailways.gov.pk

© 2023 Pakistan Railways. All rights reserved.
Serving the nation since 1861. Experience the journey of a lifetime with Pakistan Railways.



Issues Addressed:

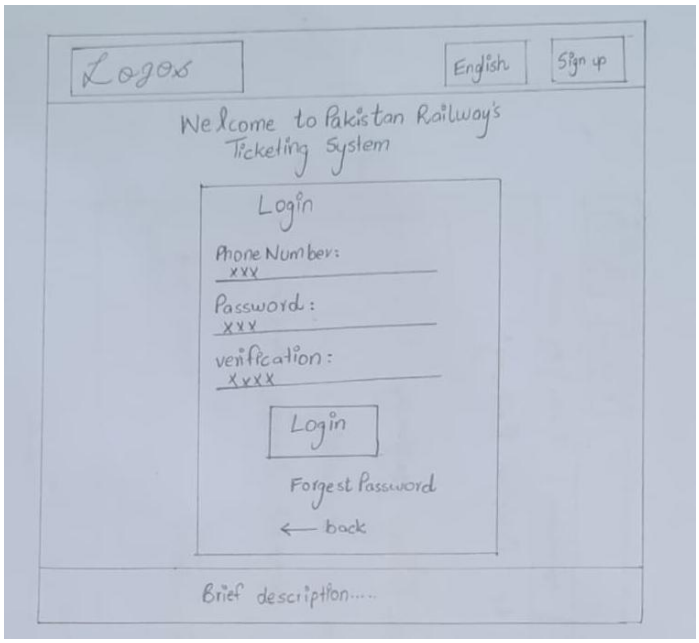
- **Poor Visual Hierarchy & Contrast:** The page now has a clear visual hierarchy, with distinct separation between the header, navigation, and main content areas.
- **Ambiguous Navigation Labels:** Navigation labels have been updated to be descriptive and intuitive e.g., "Home," "Train Timings," "Tickets," "My Account".
- **Limited Linguistic Localization:** A language selector has been added to support Urdu, Sindhi, Punjabi, and Pashto.
- **Misplaced Legal Information:** Links to Terms of Service and Privacy Policy are placed in a standard, accessible location.

Key Improvements:

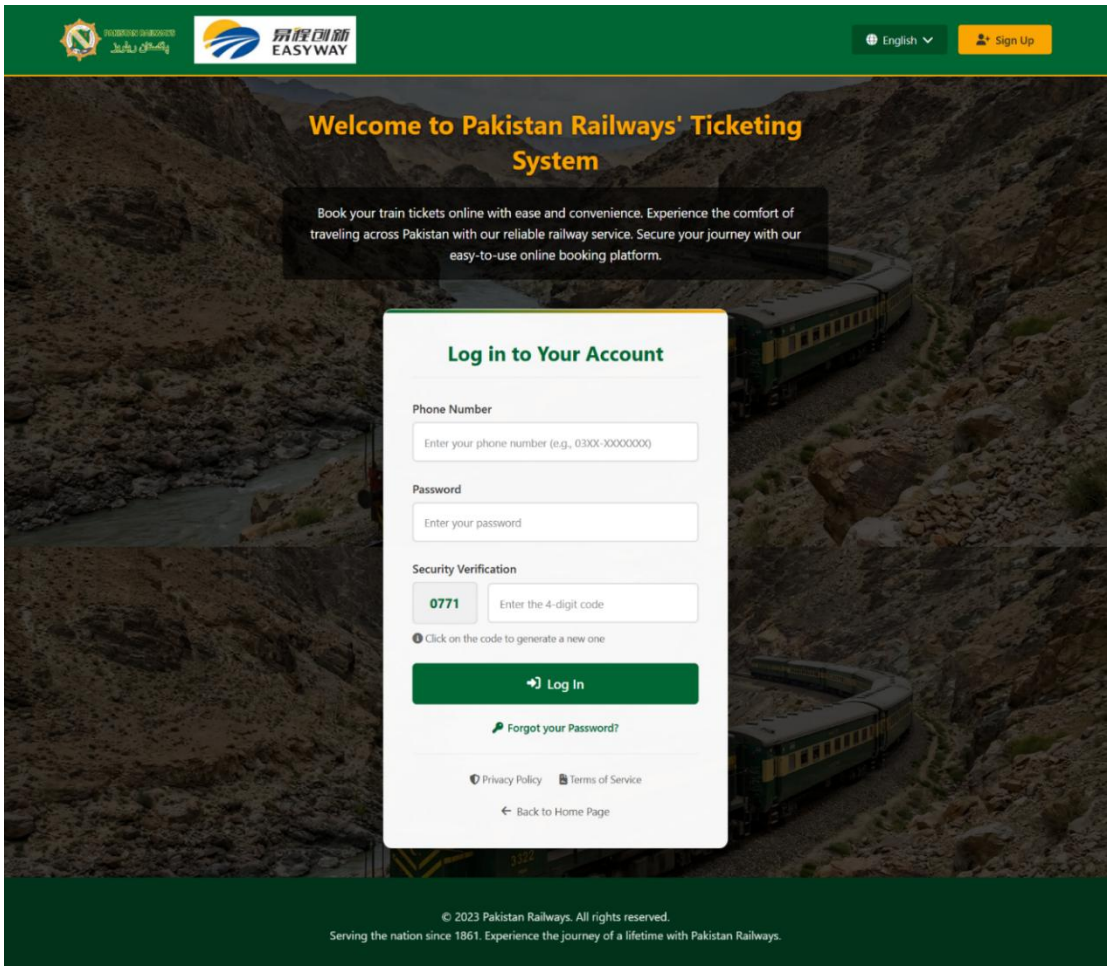
- Enhanced search form with better contrast, clear labels, and a responsive grid layout.
- Interactive elements with clear visual feedback and multiple Cultural languages support.
- Primary and secondary call-to-action buttons are clearly defined with appropriate colour coding.

ii. **Login Page:**

Lo-Fidelity:



Hi-Fidelity:



Issues Addressed:

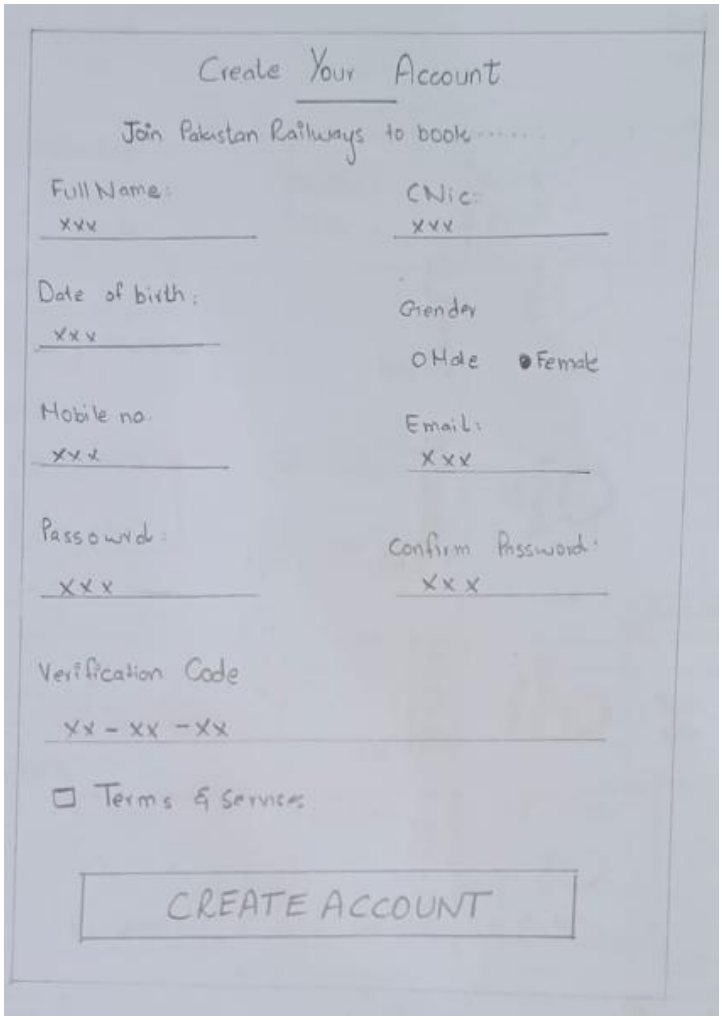
- **Poor Visibility and Contrast:** Implemented a high-contrast form with clearly distinguishable input fields.
- **Critical Functionality:** Added a clear verification code display to prevent login errors and terms "Privacy Policy" and "Terms of Service" are moved to correct place.

Key Improvements:

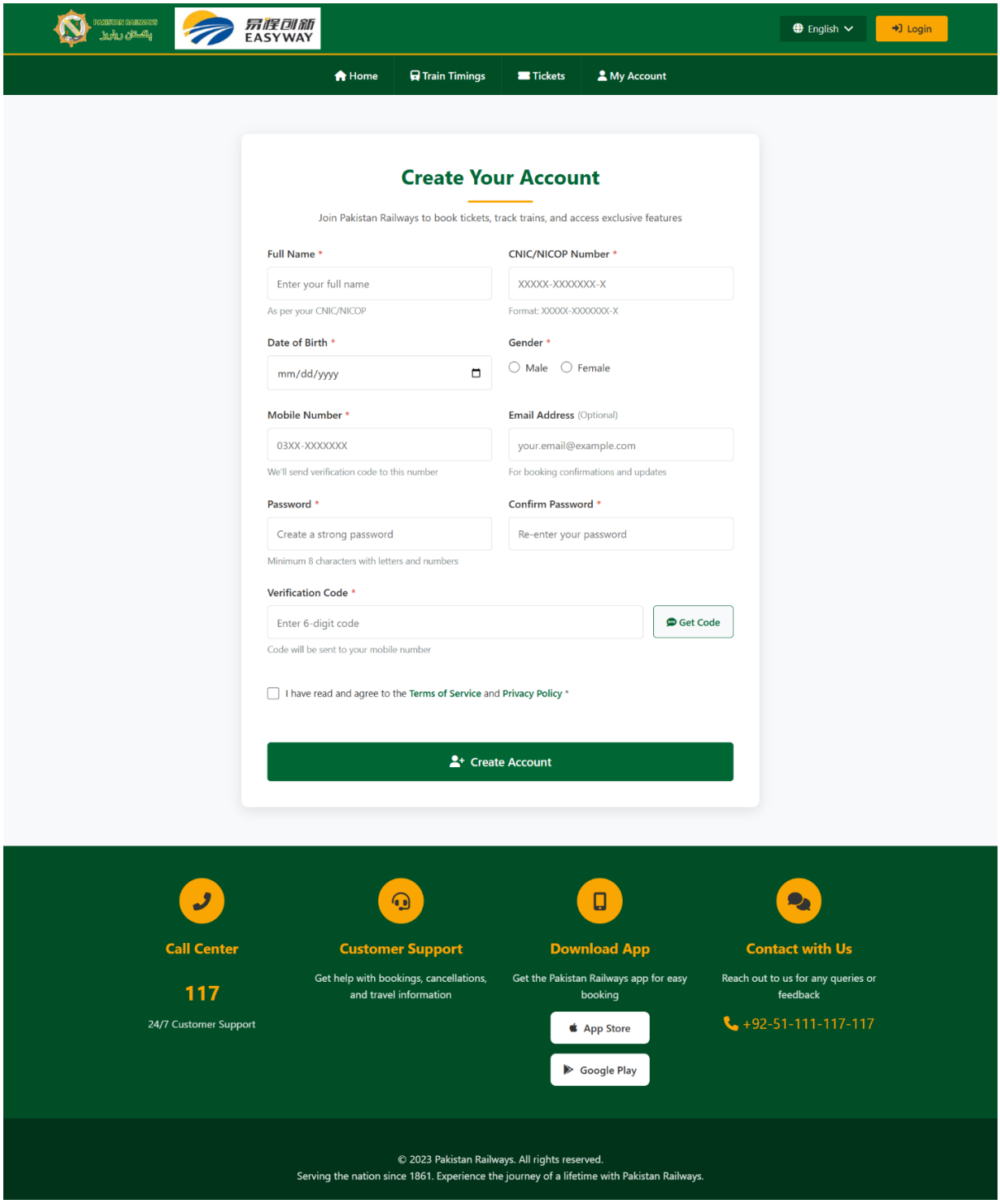
- Card-based layout with proper visual separation.
- Security code with visual feedback and regeneration option.
- Terms placed to their acquired positions.

iii. **Signup Page:**

Lo-Fidelity:



Hi-Fidelity:



Issues Addressed:

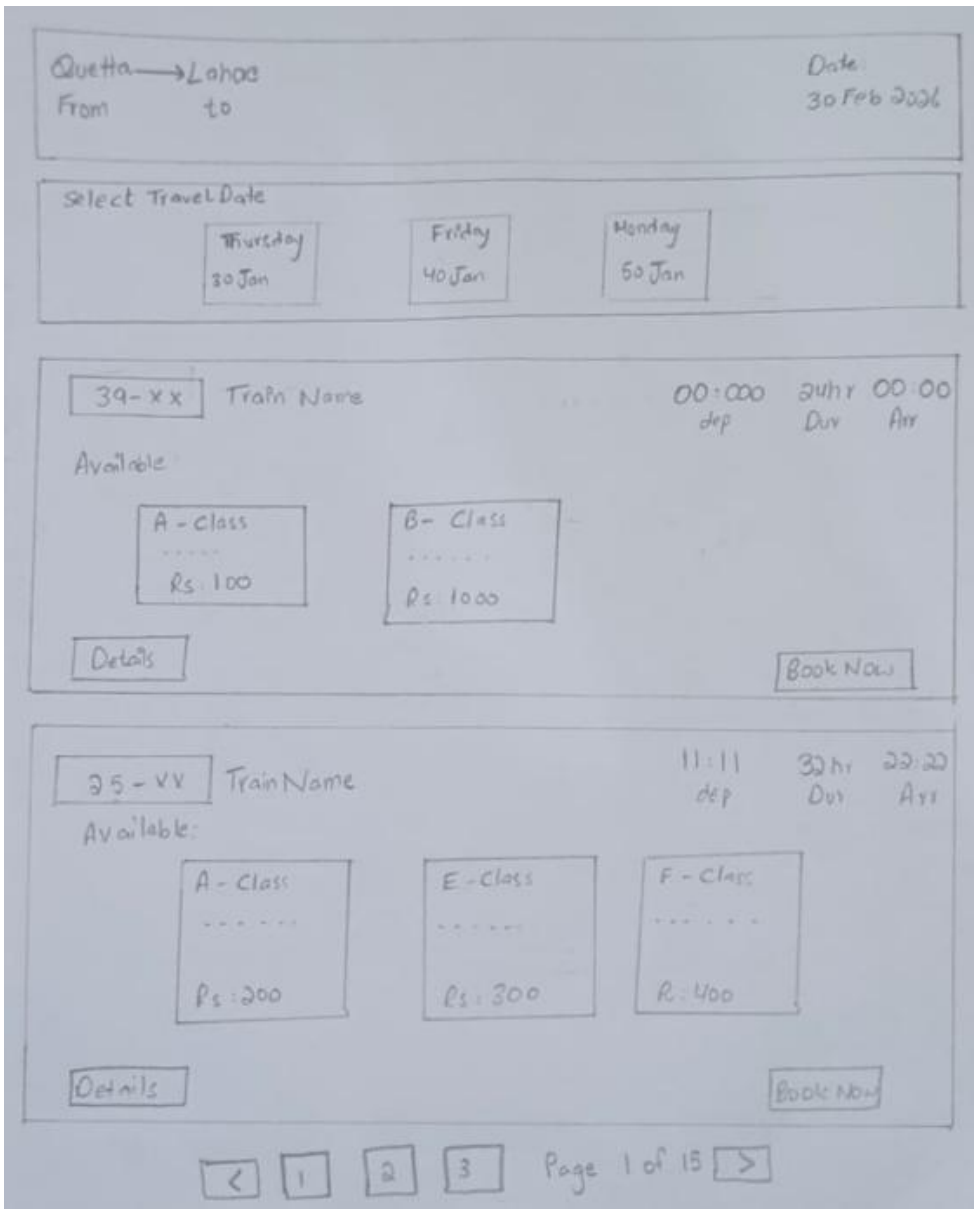
- **Spelling Errors:** Corrected "Emile Address" to "Email Address."
- **Critical Functionality Failure:** Streamlined the verification process with a clear code flow and countdown timer to prevent "Invalid Code" errors.
- **Error Prevention:** Enhanced form with real-time validation and password strength indicators.

Key Improvements:

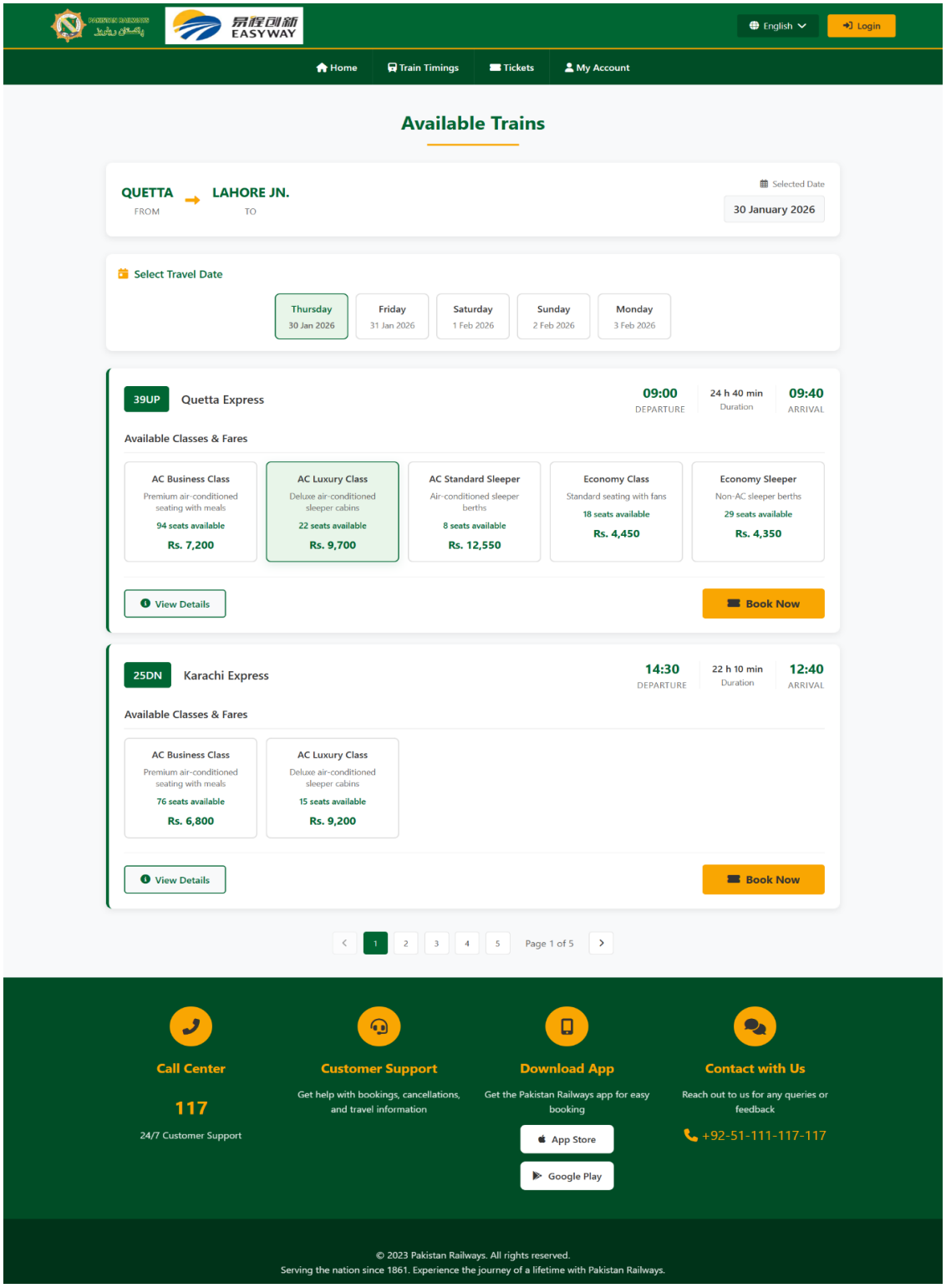
- ➔ Improved two-column form layout for better organization.
- ➔ Auto-formatting for CNIC and phone numbers.
- ➔ Clear placement of terms and conditions links.

iv. **Train Timings Page:**

Lo-Fidelity:



Hi-Fidelity:



Issues Addressed:

- **Use of Internal Jargon:** Replaced abbreviations ACSB, ACLZ with clear, user-friendly class names e.g., "AC Business Class".
- **Hidden Primary Action Button:** Made the "Book Now" button prominent and always visible, eliminating the need for horizontal scrolling.

Key Improvements:

- ➔ Improved information architecture with a card-based design for train details.
- ➔ Visual date picker with a clear selected state.
- ➔ Responsive grid for class selection and clear pricing display.

5.3 Usability Principles Applied:

Usability Principle	Original Issue in RABTA	How We Fixed It in the Redesign
Visibility of System Status	Login fields were hard to see because the colours blended into the background.	We applied high-contrast colours, clear borders, and visual feedback so users can always tell what is happening on the screen.
Match Between System and the Real World	Railway-specific codes like ACSB and ACLZ were used without explanation.	These codes were replaced with plain-language labels such as "AC Business Class" and "Economy Sleeper" so users can easily understand them.
Error Prevention	Verification often failed even when correct codes were entered.	Real-time validation, clear hints, and helpful error messages were added to prevent mistakes before they occur.
Recognition Rather Than Recall	Menu labels like "My" and "Order" were vague and confusing.	Clear, descriptive labels with icons were introduced, such as Home, Train Timings, Tickets, and My Account.
Consistency and Standards	Privacy Policy and Terms of Service were placed in unusual locations.	Legal links were standardized in the footer across all pages for consistency and easy access.
Aesthetic and Minimalist Design	Layouts were cluttered with poor spacing and grouping.	The design was cleaned up with better spacing, logical grouping, and only essential information displayed.
Flexibility and Efficiency	The "Book" button was sometimes hidden off-screen.	Primary actions like "Book Now" and "Search" were made clearly visible and easy to access on all screens.
Help Users Recognize, Diagnose, Recover from Errors	Error messages like "Invalid Code" were unhelpful and unclear.	Messages were rewritten to be specific, friendly, and guide users on what steps to take next.
Help and Documentation	There was no guidance for first-time users.	Tooltips, hints, and step-by-step instructions were added where necessary to help users navigate the system.
User Control and Freedom	Some processes felt rigid with no way to go back.	Clear "Back" buttons and cancel options were added so users always feel in control of their actions.

6. Discussion:

6.1 Comparison of Original and Redesigned Interfaces:

The heuristic evaluation uncovered several major usability issues in the original RABTA system. These issues were carefully addressed in the redesign, with focused improvements made to the interface. A side-by-side comparison highlights significant and noticeable enhancements across all the key areas that were evaluated.

Visual Design and Accessibility:

The original interface had low contrast, particularly on the login page, where input fields often blended into the background. In the redesign, the colour scheme was updated, creating a clear visual hierarchy with sufficient spacing and high-contrast interactive elements. These changes make the system much more accessible for elderly users and those with visual impairments, reducing eye strain and helping users complete tasks more efficiently.

Language and Terminology:

In the original system, internal railway abbreviations like ACSB, ACLZ, and PC were used without explanation, which often confused users. The redesigned interface replaces these with plain-language descriptions, such as *"*AC Business Class" and *"*Economy Sleeper", along with brief clarifications. This makes the system's language more intuitive, reduces cognitive load, and helps prevent booking errors caused by unfamiliar terminology.

Navigation and Information Architecture:

In the original interface, vague navigation labels like "My" and "Order" often left users guessing their purpose. In the redesign, these labels have been replaced with clear, descriptive names accompanied by icons, such as Home, Train Timings, Tickets, and My Account, following recognition-over-recall principles. This restructuring creates intuitive navigation paths, guiding users naturally through the booking process.

Error Prevention and Recovery:

The original verification process was unreliable, often rejecting valid codes and appearing at inconsistent stages. In the redesign, a robust verification flow has been implemented, featuring real-time validation, clear error messages, visual feedback, and an option for users to regenerate codes. These improvements turn a frustrating obstacle into a predictable and user-friendly process.

Cultural and Linguistic Localization:

The original interface only supported English and Urdu, limiting accessibility for many users. The redesign adds regional language support, including Punjabi, Sindhi, and Pashto, through an easy-to-use language selector. This change recognizes Pakistan's linguistic diversity and makes the system more inclusive for users who do not speak Urdu.

Consistency and Standards:

In the original system, legal information such as the Terms of Service and Privacy Policy was placed in non-standard locations, making it hard for users to find. In the redesign, these links have been consistently positioned in the footer across all pages. This follows common web conventions and helps build user trust by providing transparent access to important information.

6.2 Summary:

The heuristic evaluation proved effective in identifying major usability barriers in the original RABTA system, allowing a focused redesign that greatly improves accessibility, clarity, and efficiency. The updated interface addresses all key issues from visual contrast and terminology to navigation and error handling resulting in a system that is more inclusive, intuitive, and trustworthy. While the heuristic method successfully highlighted interface-level problems, combining it with user testing could further validate these improvements and ensure they meet the needs of Pakistan Railways' diverse user base. This demonstrates how a structured usability evaluation can drive meaningful improvements in public digital services.